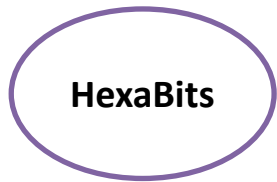


TITLE PAGE

- **Problem Statement ID** – SIH26171
- **Problem Statement Title** – On-device Visual Perception for Light-weight Browser Agents
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** –
- **Team Name** – HexaBits





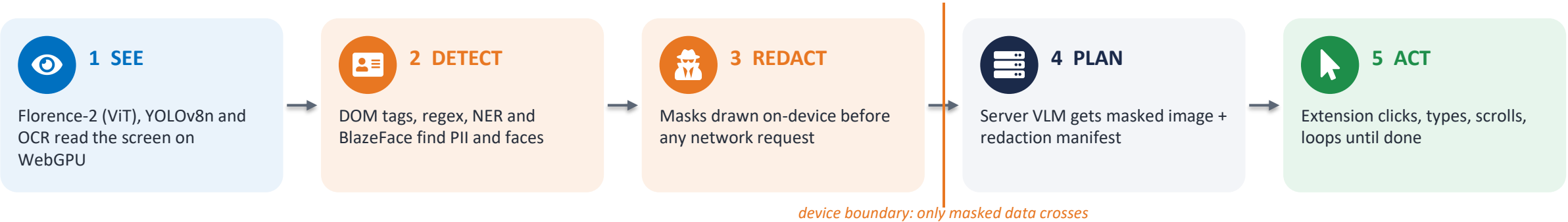
FUSION PRIVACY AGENT

See locally. Redact locally. Only then ask the server.



Proposed Solution (Describe your Idea/Solution/Prototype)

A browser extension that reads the screen with on-device vision models, masks every sensitive region locally, and lets a server VLM plan actions from the masked view only.



How it addresses the problem

- **Sensitive pixels never leave the browser:** passwords, IDs, faces and PII are masked on-device first.
- **Server stays useful:** the redaction manifest tells it what kind of field sits where, so it can still plan.
- **Light enough for a laptop:** quantized models, WebGPU with WASM fallback, all bundled offline.
- **End-to-end:** a multi-step agent loop completes real form tasks, with user confirmation on submit or pay.

Innovation and uniqueness of the solution



Technologies to be used

On-device vision / NLP

Florence-2 (ViT)

YOLOv8n

BlazeFace

BERT-NER

Tesseract OCR

Whisper-tiny

In-browser runtime

WebGPU

WASM fallback

ONNX Runtime Web

Transformers.js

MediaPipe

Extension

TypeScript

Manifest V3

Offscreen document

Side panel UI

Vite

Server

Python

FastAPI

Open-weights VLM

OpenAI-compatible API

Pydantic

Methodology and process for implementation

USER'S BROWSER (trusted, on-device)

1. Capture

Screenshot + DOM skeleton (tags, types, labels; no values)

2. Perceive

Offscreen doc: Florence-2, YOLO, OCR, NER, BlazeFace on WebGPU

3. Fuse + Redact

DOM + vision boxes, noisy-OR scoring, black-box masks

6. Risk gate + vault

Submit / pay / delete wait for user OK; asked values filled locally

7. Execute + loop

Click, type, scroll, navigate; observe again until done (max 15)

sanitized
masked image
manifest
DOM skeleton

action JSON

privacy boundary

SERVER (sees only sanitized context)

5b. Safety guard

Rejects typing into masked fields and made-up selectors

5a. VLM planner

Open-weights VLM, prompt knows the redaction scheme

4. /next-action

FastAPI receives masked PNG + manifest + DOM skeleton

Analysis of the feasibility of the idea

25%

Visual context accuracy

Florence-2 captions + YOLO elements fused with the real DOM list, so actions hit real selectors

20%

PII recall / precision

4 detectors (DOM, regex with Luhn, NER, BlazeFace) fused; eval harness: 5/5 fields, 0 false positives

20%

Redaction precision

Pixel-exact DOM boxes; DOM-safe zones suppress vision false positives

20%

Client resources

int8 / fp16 quantized models, loaded lazily in one offscreen document

15%

End-to-end latency

DOM-first fast path, WebGPU inference, fusion step under 1 ms

Potential challenges and risks

**Missed PII (false negative)****Over-redaction hides context****Heavy models, low-end laptops****Planner hallucinates actions****Site CSP blocks ML workers**

Strategies for overcoming



Fail-safe: a low-confidence lone hit is still masked



Manifest keeps field type; DOM-safe zones



Quantized, lazy load, WASM fallback



Selectors only from real DOM; server guard



ML runs in extension's offscreen document

IMPACT AND BENEFITS

Makes AI agents usable exactly where privacy matters most.

0

raw field values or PII pixels sent to the server

6

on-device models bundled in the extension, no CDN at runtime

< 1 ms

to fuse all detections into the redaction plan

Potential impact on the target audience



Citizens on e-governance portals

Agent helps fill Aadhaar, PAN, bank and scholarship forms while ID numbers stay on the device.



Government and enterprise staff

Cloud AI on internal portals without exposing confidential screens or employee data.



Low-literacy and differently-abled users

On-device voice input (Whisper) and a side panel make web tasks hands-free.

Benefits of the solution (social, economic, environmental, etc.)



Social

Trust in AI help; data minimisation in line with the DPDP Act 2023.



Economic

Perception runs on the client; open-weights server model, no vendor lock-in.



Security

Server never stores raw PII, so a server breach leaks nothing personal.



Environmental

Less server GPU work and smaller payloads per step.

Details / Links of the reference and research work



On-device models

Florence-2 (ViT)

Xiao et al., CVPR 2024, arXiv:2311.06242

BlazeFace

Bazarevsky et al., arXiv:1907.05047

YOLOv8

github.com/ultralytics/ultralytics

BERT NER

Devlin et al., arXiv:1810.04805; hf.co/Xenova/bert-base-NER

Whisper

Radford et al., arXiv:2212.04356

Tesseract OCR

github.com/naptha/tesseract.js



Runtimes and platform

ONNX Runtime Web

onnxruntime.ai/docs/tutorials/web

Transformers.js

huggingface.co/docs/transformers.js

WebGPU

w3.org/TR/webgpu

MediaPipe Tasks

ai.google.dev/edge/mediapipe

Chrome Offscreen API

developer.chrome.com/docs/extensions

FastAPI

fastapi.tiangolo.com



Web agents and policy

SeeAct

Zheng et al., ICML 2024, arXiv:2401.01614

Mind2Web

Deng et al., NeurIPS 2023, arXiv:2306.06070

Llama 3.2 Vision

hf.co/meta-llama/Llama-3.2-11B-Vision-Instruct

DPDP Act 2023

meity.gov.in, Digital Personal Data Protection Act

SIH 2026 PS 26171

ISRO / SAC, On-device Visual Perception for Browser Agents



Our own research: working prototype + reproducible eval harness

MV3 extension + FastAPI server built and running. Eval (eval/results.md) runs the real redaction code on a labelled fixture: 5/5 sensitive fields redacted, 0/2 false positives, 0 Luhn-invalid card hits, fusion in 0.85 ms.